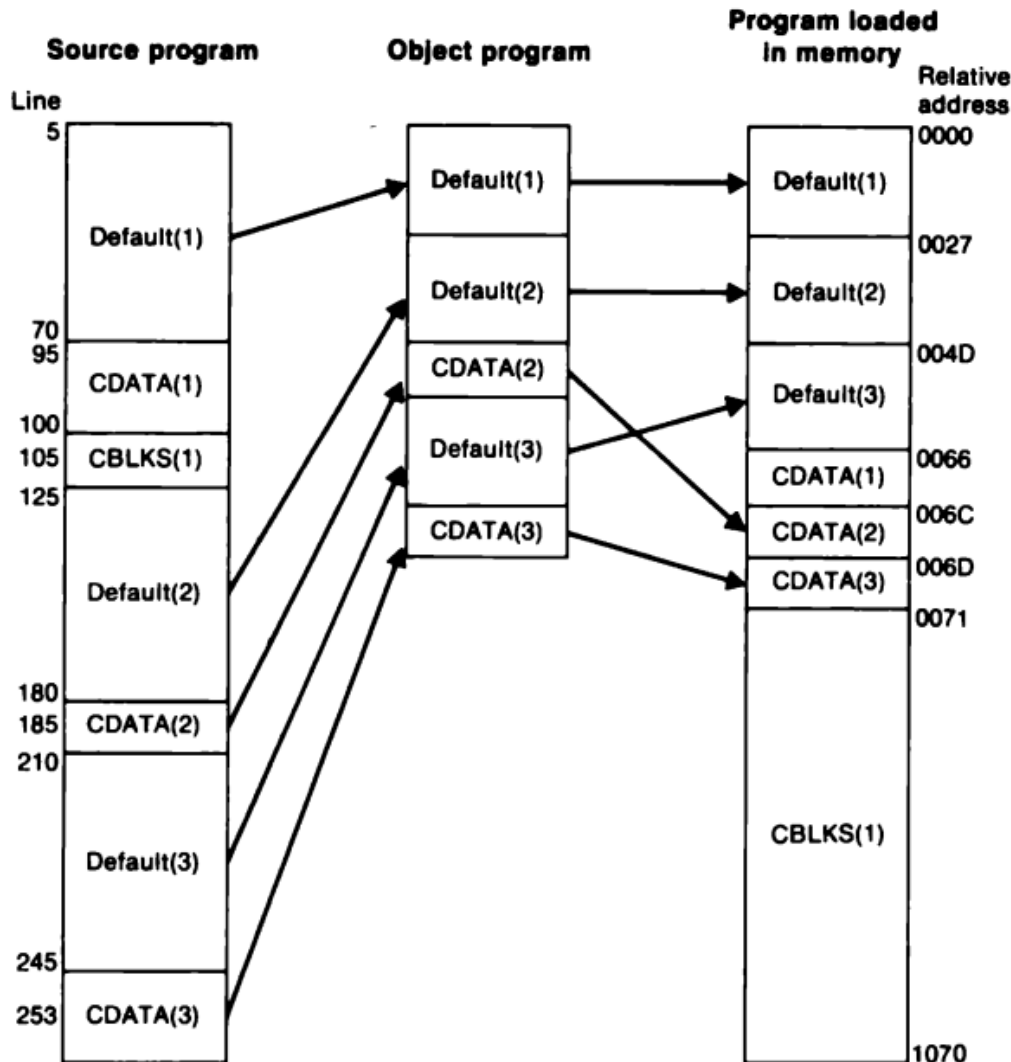


# Section 8

Systems Programming ECE3502

# Program Blocks



The assembler directive "**USE**" indicates which portions of the source program belong to the various blocks

We will divide our program to 3 blocks:

- **Unnamed/ default block** contains all the executable instruction.
- **CBLKS** block contains all data areas that are a few word.
- **CBLKS** block contains all data areas that of larger blocks of the memory.

# Book, Page 81: Pass One

| Location/Blocks | Symbols | Instructions | reference      |
|-----------------|---------|--------------|----------------|
| 0000 0          | COPY    | START        | 0              |
| 0000 0          | FIRST   | STL          | RETADR         |
| 0003 0          | CLOOP   | JSUB         | RDREC          |
| 0006 0          |         | LDA          | LENGTH         |
| 0009 0          |         | COMP         | #0             |
| 000C 0          |         | JEQ          | ENDFIL         |
| 000F 0          |         | JSUB         | WRREC          |
| 0012 0          |         | J            | CLOOP          |
| 0015 0          | ENDFIL  | LDA          | =C' EOF'       |
| 0018 0          |         | STA          | BUFFER         |
| 001B 0          |         | LDA          | #3             |
| 001E 0          |         | STA          | LENGTH         |
| 0021 0          |         | JSUB         | WRREC          |
| 0024 0          |         | J            | @RETADR        |
| 0000 1          |         | USE          | CDATA          |
| 0000 1          | RETADR  | RESW         | 1              |
| 0003 1          | LENGTH  | RESW         | 1              |
| 0000 2          |         | USE          | CBLKS          |
| 0000 2          | BUFFER  | RESB         | 4096           |
| 1000 2          | BUFEND  | EQU          | *              |
| 1000 --         | MAXLEN  | EQU          | BUFEND-BUFFER. |

| Location/Blocks | Symbols | Instructions | reference |
|-----------------|---------|--------------|-----------|
| 0027 0          |         | USE          |           |
| 0027 0          | RDREC   | CLEAR        | X         |
| 0029 0          |         | CLEAR        | A         |
| 002B 0          |         | CLEAR        | S         |
| 002D 0          |         | +LDT         | #MAXLEN   |
| 0031 0          | RLOOP   | TD           | INPUT     |
| 0034 0          |         | JEQ          | RLOOP     |
| 0037 0          |         | RD           | INPUT     |
| 003A 0          |         | COMPR        | A, S      |
| 003C 0          |         | JEQ          | EXIT      |
| 003F 0          |         | STCH         | BUFFER, X |
| 0042 0          |         | TIXR         | T         |
| 0044 0          |         | JLT          | RLOOP     |
| 0047 0          | EXIT    | STX          | LENGTH    |
| 004A 0          |         | RSUB         |           |
| 0006 1          |         | USE          | CDATA     |
| 0006 1          | INPUT   | BYTE         | X' F1'    |

There is no block number for MAXLEN. This is because MAXLEN is an absolute symbol.

# Program continue

| Location/Blocks |   | Symbols | Instructions | reference |
|-----------------|---|---------|--------------|-----------|
| 004D            | 0 |         | USE          |           |
| 004D            | 0 | WRREC   | CLEAR        | X         |
| 004F            | 0 |         | LDT          | LENGTH    |
| 0052            | 0 | WLOOP   | TD           | =X'05'    |
| 0055            | 0 |         | JEQ          | WLOOP     |
| 0058            | 0 |         | LDCH         | BUFFER,X  |
| 005B            | 0 |         | WD           | =X'05'    |
| 005E            | 0 |         | TIXR         | T         |
| 0060            | 0 |         | JLT          | WLOOP     |
| 0063            | 0 |         | RSUB         |           |
| 0007            | 1 |         | USE          | CDATA     |
|                 |   |         | LTORG        |           |
| 0007            | 1 | *       | =C'EOF'      |           |
| 000A            | 1 | *       | =X'05'       |           |
|                 |   |         | END          | FIRST     |

| Block name | Block number | Address | Length |
|------------|--------------|---------|--------|
| (Default)  | 0            | 0000    | 0066   |
| CDATA      | 1            | 0066    | 000B   |
| CBLKS      | 2            | 0071    | 1000   |

| Symbol | Type | Value |
|--------|------|-------|
| FIRST  | R    | 0000  |
| CLOOP  | R    | 0003  |
| ENDFIL | R    | 0015  |
| RETADR | R    | 0066  |
| LENGHT | R    | 0069  |
| BUFFER | R    | 0071  |
| BUFEND | R    | 1071  |
| MAXLEN | A    | 1000  |
| RDREC  | R    | 0027  |
| RLOOP  | R    | 0031  |
| EXIT   | R    | 0047  |
| INPUT  | R    | 006C  |
| WRREC  | R    | 004D  |
| WLOOP  | R    | 0052  |

| Literal name | Value  | Length | Address |
|--------------|--------|--------|---------|
| =C'EOF'      | 454F46 | 3      | 006D    |
| =X'05'       | 05     | 1      | 0070    |

# Book, Page 83 : Pass Two

| Location/Blocks |   | Symbols | Instructions | reference     | Obj. code    |
|-----------------|---|---------|--------------|---------------|--------------|
| 0000            | 0 | COPY    | START        | 0             |              |
| 0000            | 0 | FIRST   | STL          | RETADR        | 172063       |
| 0003            | 0 | CLOOP   | JSUB         | RDREC         | 4B2021       |
| 0006            | 0 |         | LDA          | LENGTH        | 032060       |
| 0009            | 0 |         | COMP         | #0            | 290000       |
| 000C            | 0 |         | JEQ          | ENDFIL        | 332006       |
| 000F            | 0 |         | JSUB         | WRREC         | 4B203B       |
| 0012            | 0 |         | J            | CLOOP         | 3F2FEE       |
| 0015            | 0 | ENDFIL  | LDA          | =C' EOF'      | 032055       |
| 0018            | 0 |         | STA          | BUFFER        | 0F2056       |
| 001B            | 0 |         | LDA          | #3            | 010003       |
| 001E            | 0 |         | STA          | LENGTH        | 0F2048       |
| 0021            | 0 |         | JSUB         | WRREC         | 4B2029       |
| 0024            | 0 |         | J            | @RETADR       | 3E203F       |
| 0000            | 1 |         | USE          | CDATA         | No Obj. code |
| 0000            | 1 | RETADR  | RESW         | 1             | No Obj. code |
| 0003            | 1 | LENGTH  | RESW         | 1             | No Obj. code |
| 0000            | 2 |         | USE          | CBLKS         | No Obj. code |
| 0000            | 2 | BUFFER  | RESB         | 4096          | No Obj. code |
| 1000            | 2 | BUFEND  | EQU          | *             | No Obj. code |
| 1000            | 2 | MAXLEN  | EQU          | BUFEND-BUFFER | No Obj. code |

## STL RETADR

| Opcode<br>(6) | n  | i | x | b    | p | e | Disp(12)<br>(TA - PC) |
|---------------|----|---|---|------|---|---|-----------------------|
| 14            | 1  | 1 | 0 | 0    | 1 | 0 | (000+066) - 003= 063  |
| 0001 01       | 11 |   |   | 0010 |   |   | 0000 0110 0011        |

**172063**

## JSUB RDREC

| Opcode<br>(6) | n  | i | x | b    | p | e | Disp(12)<br>(TA - PC) |
|---------------|----|---|---|------|---|---|-----------------------|
| 48            | 1  | 1 | 0 | 0    | 1 | 0 | 027- 006= 021         |
| 0100 10       | 11 |   |   | 0010 |   |   | 0000 0010 0001        |

**4B2021**

## J @RETADR

| Opcode<br>(6) | n  | i | x | b    | p | e | Disp(12)<br>(TA - PC) |
|---------------|----|---|---|------|---|---|-----------------------|
| 3C            | 1  | 0 | 0 | 0    | 1 | 0 | (000+066)- 0027= 03F  |
| 0011 11       | 10 |   |   | 0010 |   |   | 0000 0011 1111        |

**3E203F**

# Book, Page 83 : Pass Two

| Location/Blocks |   | Symbols | Instructions | reference | Obj. code    |
|-----------------|---|---------|--------------|-----------|--------------|
| 0027            | 0 |         | USE          |           |              |
| 0027            | 0 | RDREC   | CLEAR        | X         | B410         |
| 0029            | 0 |         | CLEAR        | A         | B400         |
| 002B            | 0 |         | CLEAR        | S         | B440         |
| 002D            | 0 |         | +LDT         | #MAXLEN   | 75101000     |
| 0031            | 0 | RLOOP   | TD           | INPUT     | E32038       |
| 0034            | 0 |         | JEQ          | RLOOP     | 332FFA       |
| 0037            | 0 |         | RD           | INPUT     | DB2032       |
| 003A            | 0 |         | COMPR        | A, S      | A004         |
| 003C            | 0 |         | JEQ          | EXIT      | 332008       |
| 003F            | 0 |         | STCH         | BUFFER, X | 57A02F       |
| 0042            | 0 |         | TIXR         | T         | B850         |
| 0044            | 0 |         | JLT          | RLOOP     | 3B2FEA       |
| 0047            | 0 | EXIT    | STX          | LENGTH    | 13201F       |
| 004A            | 0 |         | RSUB         |           | 4F0000       |
| 0006            | 1 |         | USE          | CDATA     | No Obj. code |
| 0006            | 1 | INPUT   | BYTE         | X'F1'     | F1           |

## CLEAR X

| Opcode(8) | R1(4) | R2(4) |
|-----------|-------|-------|
| B4        | 1     | 0     |
| 1011 0100 | 0001  | 0000  |

**B410**

## +LDT #MAXLEN

| Opcode<br>(6) | n  | i | x    | b | p | e | Addr(20)                 |
|---------------|----|---|------|---|---|---|--------------------------|
| 74            | 0  | 1 | 0    | 0 | 0 | 1 | 01000                    |
| 0111 01       | 01 |   | 0001 |   |   |   | 0000 0001 0000 0000 0000 |

**75101000**

## STCH BUFFER, X

| Opcode<br>(6) | n  | i | x    | b | p | e | Disp(12)<br>(TA – PC) |
|---------------|----|---|------|---|---|---|-----------------------|
| 54            | 1  | 1 | 1    | 0 | 1 | 0 | (000+071)– 042= 02F   |
| 0101 01       | 11 |   | 1010 |   |   |   | 0000 0010 1111        |

**57A02F**



# Book, Page 83 : Pass Two

| Location/Blocks |   | Symbols | Instructions | reference | Obj. code    |
|-----------------|---|---------|--------------|-----------|--------------|
| 004D            | 0 |         | USE          |           |              |
| 004D            | 0 | WRREC   | CLEAR        | X         | B410         |
| 004F            | 0 |         | LDT          | LENGTH    | 772017       |
| 0052            | 0 | WLOOP   | TD           | =X'05'    | E3201B       |
| 0055            | 0 |         | JEQ          | WLOOP     | 332FFA       |
| 0058            | 0 |         | LDCH         | BUFFER,X  | 53A016       |
| 005B            | 0 |         | WD           | =X'05'    | DF2012       |
| 005E            | 0 |         | TIXR         | T         | B850         |
| 0060            | 0 |         | JLT          | WLOOP     | 3B2FEF       |
| 0063            | 0 |         | RSUB         |           | 4F0000       |
| 0007            | 1 |         | USE          | CDATA     | No Obj. code |
|                 |   |         | LTORG        |           | No Obj. code |
| 0007            | 1 | *       | =C'EOF       |           | 454F46       |
| 000A            | 1 | *       | =X'05'       |           | 05           |
|                 |   |         | END          | FIRST     |              |

H. program name(6). Starting address(6). Length(6)  
H.COPYXX.000000.001071

T. Starting address of each t record(6) . Size of instruction(2). obj. code of all inst(6)

T.000000.1E.172063.4B2021.032060.290000.332006  
.4B203B.3F2FEE.032055.0F2056.010003

T.00001E.09.0F2048.4B2029.3E203F

T.000027.1D.B410.B400.B440.75101000.E32038.332  
FFA.DB2032.A004.332008.57A02F.B850

T.000044.09.3B2FEA.13201F.4F000

T.00006C.01.F1

T.00004D.19.B410.772017.E3201B.332FFA.53A016.  
DF2012.B850.3B2FEF.4F0000

T.00006D.04.454F46.05

E. Starting address(6)

E.000000

# NOTES

- `+LDT #MAXLEN`

NO need for M record???

Since MAXLEN is an absolute address => `MAXLEN EQU BUFFEND-BUFFER`

- **Absolute symbols** (e.g. MAXLEN in the previous example) are **constant values**, not a memory location.

They are stored in the **assembler's symbol table**, not in memory.

Thus, **no block number** is assigned.